

# Individualized Generation of Upper Limb Training for Robot-assisted Rehabilitation using Multi-objective Optimization

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## ABSTRACT

To alleviate the labor costs required to customizing the content and difficulty of the rehabilitation game for each patient, we propose a method to automatically generate individualized training contents on desktop end-effector rehabilitation robot. By modeling the search of the training motions as finding optimal hand paths and trajectories, we introduce solving the design problem with a multi-objective optimization (MO) solver. Our system is capable of automatically generating various training plans considering the training intensity and dexterity of each joint in the upper limb. In addition, we develop a serious game to display the generated training, which helps motivate the patient in the rehabilitation.

**Index Terms:** Applied computing—Life and medical sciences—Health informatics—; Human-centered computing—Interaction design—Interaction design process and methods—User centered design

## 1 INTRODUCTION

In the past two decades, robot-assisted rehabilitation as an emerging treatment approach has been increasingly used in neurological central nervous system conditions. It performs highly repetitive, intensive, task oriented and quantifiable therapies for motor rehabilitation with precise measurements and objective assessment [5]. However, robot-assisted approaches still rely heavily on therapists' subjective clinical decisions on the recovery plan. If the training content can be well specified to the robots automatically, the upper limb rehabilitation will be greatly boosted by robot-assisted rehabilitation.

In recent years, generation based approaches are proposed to achieve automatic customization for rehabilitation, such as the task generator that creates personalized task for cognitive rehabilitation [3], and the fuzzy-logic-based autonomous exercise generation system [6]. However, a common limitation of the above methods is that they can only retrieve for a training plan from an existing library, but cannot synthesis the plan from a continuous motion space.

In addition, serious game has been proved its effectiveness as well as the patient's immersion and involvement [4]. We plan to automate the content synthesis for typical serious games used in robot-assisted rehabilitation.

## 2 HARDWARE SETTING

We use a desktop force-feedback rehabilitation system for upper limb. It has two degrees of freedom to produce planar translational motions, supporting active and passive rehabilitation training modes. The robot implements a hybrid linkage structure to transmit the rotational motion of the motors to the planar translation of the end-effector. An arm bracket is linked to the end effector to support the affected arm. It allows a patient sitting in front of the robot to grasp the end-effector. The patient can actively move or be passively

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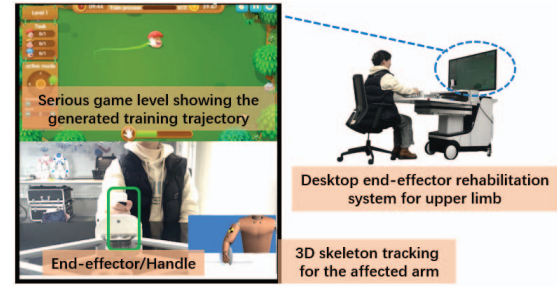


Figure 1: Overview of our system.

dragged by the end-effector according to the instructions on the screen so that the shoulder and elbow joints are trained with the upper-limb motions. Whatever the training modality is, an exercise plan is needed to tell the patient where to move next. Therefore, our goal is to recommend a trajectory to lead the patient's movement.

## 3 PROTOTYPE DESIGN

### 3.1 Difficulty Quantification

We decouple the concept of difficulty into the intensity and the dexterity [1]. Exercise intensity is used to describe how much physical effort will be taken to complete the exercises within the allotted time. Dexterity is adroitness in using body and describes the degree to which the training includes fine and sensitive movements. In our case, when the end effector performs planar translational motion with user control, the shape of the path is directly related to intensity by acting on the rotation angles of each joint (the shoulder joint and the elbow joint). Meanwhile, dexterity is strength-related since there must be enough force control to perform the tasks of dexterity. Therefore, we model the dexterity using the trajectory with the angular velocity and acceleration of joints, as the differential quantities are closely related to the force exerted in the training.

### 3.2 Two-stage Training Generator

Given the desired difficulty specified by the therapist as goals, combined with the patient information collected in advance as constraints, we aim to generate a group of individual-adapted exercises over the desktop robotic platform. Based on our definition of difficulty, we further divide our goal into two sub-problems: path generation and trajectory planning. Path generation is to generate a pure geometric path, then the trajectory planning takes a given geometric path and allocates velocity for each time slice.

We formulate the two stages into multi-objective optimization problems. We propose to separately solve

$$\begin{aligned} \arg \min_{\mathcal{P}} \mathbf{f}_1(\mathcal{P}) &:= (C_R^S(\mathcal{P}), C_R^E(\mathcal{P}), C_L(\mathcal{P})) \\ \text{s.t. } x_i &\in \mathcal{P} = (x_0, \dots, x_n) \in \mathcal{X} \end{aligned} \quad (1)$$

Table 1: Tracking performance of G1 and G2 (*Mean. ± Std.*)

| Group | Low           | Middle        | High          |
|-------|---------------|---------------|---------------|
| G1    | 1.5965±0.2239 | 1.9176±0.3042 | 2.1505±0.3676 |
| G2    | 1.4220±0.3110 | 1.6089±0.2186 | 1.8221±0.3186 |

and

$$\begin{aligned} \arg \min_{\mathcal{T}=(v_0, \dots, v_m)} \quad & \mathbf{f}_2(\mathcal{T}) := (C_D^S(\mathcal{T}), C_D^E(\mathcal{T})) \\ \text{s.t.} \quad & \sum_{i=0}^m v_i t / m = L(\mathcal{P}) \end{aligned} \quad (2)$$

where the Boltzmann-like cost functions  $C$  measure the difference between the actual performance and its target specified by the therapists. For path generation (Equation 1), we are to minimize optimization objectives including the costs of the length of the path  $C_L(\mathcal{P})$ , absolute rotational angle of the shoulder joint  $C_R^S(\mathcal{P})$  and of the elbow joint  $C_R^E(\mathcal{P})$ . For trajectory planning (Equation 2), we design a scoring strategy for measuring the dexterity of the joint angular velocity, which is related to the differentials of the angular velocity, and use the dexterity scoring for the shoulder joint  $C_D^S(\mathcal{T})$  and for the elbow joint  $C_D^E(\mathcal{T})$  as objectives for optimization. The integration of the velocity  $v_i$  yields to the length of the path  $L(\mathcal{P})$ . In addition, we constrain the points  $x_i$  in the path  $\mathcal{P}$  always within the motion range  $\mathcal{X}$  of the user and the velocity  $v_i$  of the trajectory  $\mathcal{T}$  is always set in a reasonable range, which is measured before using the rehabilitation system. The two-stage optimization is to find the optimal  $\mathcal{P}_{opt}$  and  $\mathcal{T}_{opt}$  that minimize the cost vectors. These multi-objective optimization problems are solved by a multi-objective optimizer *Archived Multi-Objective Simulated Annealing* (AMOSA) algorithm [2].

#### 4 USER STUDY AND RESULTS

Our experimental system is shown in Fig. 1. We developed a serious game named *Chasing the Mushroom* on the robot platform and conducted a user study on it to validate our exercise generator. When the game starts, a mushroom will move according to the generated training trajectory and the user is asked to follow its movement to earn scores by controlling the handle. Active rehabilitation mode was adopted during our study. There was no external assistance to force user movement and the user only needed to move according to his motor ability. We designed 3 training groups including Group 1 (G1): three exercises with different intensity targets and same dexterity targets; Group 2 (G2): three exercises with same intensity targets and different dexterity targets; Group 3 (G3): an elbow-focused and a shoulder-focused exercises, with same dexterity target.

We recruited 11 participants including MSc and PhD students aged from 20 to 30, having previous experience evaluating rehabilitation applications and no prior knowledge of our study. After getting familiar with the game control, subjects were asked to play the game in the 3 groups with a random order and the exercises in each group were also shown in a random order. Meanwhile, the user's real trajectories were recorded at a frequency of 20Hz and the real skeletal data of the user's upper limb was collected by Kinect for windows 2.0. This study has obtained ethics approval.

We computed the rooted mean square error (RMSE) between the target trajectory and the recorded trajectory to measure the performance of G1 and G2, as listed in Table 1. Statistically, the results of one-way repeated measure analysis of variance (ANOVA) and the paired t-test for each pair setting of intra-group indicate that the significance of the difference between every pair of difficulty settings. It is also proved that the intensity and dexterity in our model are able to describe and control the training difficulty.

We computed the actual amount of joint rotation in G3 from the collected skeleton data, as shown in Fig. 2. We found that the actual joint movement in training had overall consistent trends with

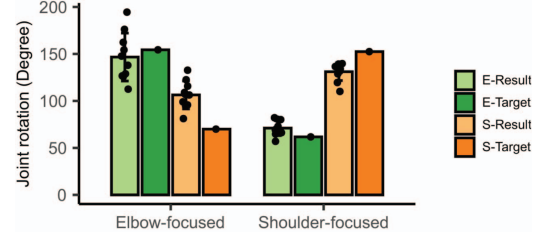


Figure 2: Participants' total joint rotations compared with the joint rotation targets used in G3. The real results were positively correlated to the corresponding targets.

the targets in elbow-focused and shoulder-focused training. We also conducted a t-test to compare the total rotation on different joints. The results indicated that the total amount of joint rotations on the elbow and shoulder had significant differences during focused training. This result benefits from our multi-objective optimizer for different joints.

#### 5 CONCLUSION AND FUTURE WORK

We propose a method to generate training plans for the desktop robot for upper limb rehabilitation. We devise a two-stage approach based on MO to generate exercises through path generation and trajectory planning. Our approach could adapt to different motor abilities and training goals from patients, and take into account the length and the rotation of each joint for training intensity, the duration and smoothness for training dexterity. By using our developed serious game to display generated exercises and conducting user experiments, we validate the feasibility of the goal-guided training generation. In the future, we plan to invite more participants with upper limb motor dysfunction and use advanced motion capture systems to avoid the limited precision of Kinect in our user study. Moreover, our task generator could be also extended to support the level design for 3D immersive VR games by setting the path and trajectory in 3D space. Next, we would like to build more immersive exergames in VR with the training plans generated from our method.

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